EMIPredict AI – Business Impact Assessment

Recommendations for financial institutions, FinTech companies, banks and loan officers

Live app: https://abhisheknimaje435-emipredict-ai-app-ccio5a.streamlit.app

Repository: https://github.com/AbhishekNimaje435/emipredict-ai

1. Executive summary

EMIPredict AI is a live, web-based risk assessment platform that answers two questions for every loan applicant: Is this applicant eligible for the EMI? (Eligible / High_Risk / Not_Eligible) and What is the maximum monthly EMI they can safely afford? (in INR). Both answers come from XGBoost models trained on about 400,000 financial profiles across five EMI scenarios and tracked in MLflow.

On the held-out test set the platform reaches 97.5% eligibility accuracy and a maximum-EMI error (RMSE) of ₹746, well inside the project targets of 90% and ₹2,000. In validation, the model's most costly error, approving someone who should be declined, occurred for only 69 of 46,378 Not_Eligible applicants (0.15%).

The central business conclusion: about 94% of applications can be decided automatically, and human underwriters can concentrate on the roughly 6% of borderline cases (details in Section 4).

2. The platform at a glance

Page	What it does	Who benefits
Prediction (single applicant)	Enter 25 applicant fields and get eligibility, confidence, class probabilities, maximum safe EMI, requested EMI and headroom, plus a financial health snapshot (debt-to-income, expense-to-income, requested EMI-to-income, disposable income, savings buffer)	Loan officers, walk-in counters
Prediction (batch CSV)	Score many applicants in one upload	Underwriting teams, FinTech back-offices

Page	What it does	Who benefits
Data Explorer	Filter by scenario, eligibility and employment type; distributions, relationships, engineered ratios, correlations, raw data	Risk and portfolio analysts
Model Performance	Model comparison table and charts, confusion matrix, feature importance for the selected models, registry details	Model-risk and compliance teams
Admin Data Management	Create, read, update and delete financial records with filtering and pagination (password-protected)	Data administrators

3. What the data shows

The Data Explorer works on a random sample of 20,000 cleaned records: 18.6% are Eligible, average monthly salary is ₹57,663 and average maximum EMI is ₹6,811. On the full cleaned dataset of 400,024 records, 18.4% are Eligible, 77.3% Not_Eligible and 4.3% High_Risk.

3.1 Eligibility differs sharply by EMI scenario

EMI scenario	Eligible (approx.)	Not_Eligible (approx.)
E-commerce Shopping	~27%	~68%
Home Appliances	~25%	~69%
Education	~18%	~77%
Personal Loan	~12%	~85%
Vehicle	~10%	~86%

The values are approximate readings from the Eligibility share by EMI Scenario chart. Smaller-ticket, shorter-tenure products have the highest eligibility; high-ticket products (vehicle, personal loan) have the lowest. This is expected, since a larger requested EMI

consumes more of the applicant's disposable income. The average maximum safe EMI itself is almost identical across scenarios (about ₹6,735 to ₹6,783), so the difference comes from the size of the request, not from the applicant's capacity.

3.2 Income matters, but it is not the whole story

Most applicants cluster between roughly ₹20,000 and ₹80,000 per month and are mostly Not_Eligible; Eligible applicants become visible mainly at higher salaries. The spike at the top of the range (about ₹175,000) comes from clipping salaries to the 1st–99th percentile during data cleaning, so it is not a real concentration of applicants. Income alone does not decide eligibility: many high earners are still Not_Eligible because of existing EMIs and expenses.

3.3 Affordability ratios drive decisions

The engineered ratios (Step 3 of the project) dominate both models, which confirms the value of feature engineering. Gender and marital status do not appear among the top 15 features of either model, but a formal fairness audit is still recommended (Section 6).

Rank	Eligibility model (classification)	Maximum-EMI model (regression)
1	EMI-to-disposable-income ratio (~0.16)	Existing loans = Yes (~0.65)
2	Existing loans, Yes/No (~0.09 each, ~0.18 combined)	Disposable income (~0.14)
3	Debt-to-income × credit score	Salary × credit score
4	EMI-to-income ratio	EMI-to-disposable ratio
5	Affordability ratio	Debt-to-income × credit score

Existing loans is encoded as two columns (Yes and No) of the same variable, so combined it carries about as much weight as the top ratio in the eligibility model.

4. Model performance in business terms

Validation confusion matrix (60,004 applicants):

Actual \ Predicted	Eligible	High_Risk	Not_Eligible	Total
Eligible	10,588	385	60	11,033
High_Risk	128	2,348	117	2,593
Not_Eligible	69	727	45,582	46,378

- Costliest error is rare: only 69 Not_Eligible applicants (0.15%) were predicted Eligible. Only 60 Eligible applicants (0.5%) were wrongly predicted Not_Eligible, so few good customers are lost.
- Errors lean cautious: most mistakes fall into High_Risk (727 Not_Eligible and 385 Eligible applicants), which sends them to review or higher pricing instead of an outright wrong decision.
- Automation potential: the model predicts High_Risk for 3,460 applicants (5.8%). If only these go to a human underwriter and the other 94.2% are decided automatically, manual workload falls by roughly 94% in this simulation. This is a projection from validation data, not a measured production result.
- High_Risk precision is about 68% (2,348 of 3,460 predicted). This is acceptable for a refer-to-underwriter bucket, but it should not be used as a hard decline.

5. Impact by stakeholder

Stakeholder	Impact
Financial institutions	Faster underwriting; one standard eligibility rule across all five EMI scenarios; risk-based pricing using the High_Risk class; instant eligibility for walk-in customers.
FinTech companies	Instant pre-qualification at checkout or in a mobile app; batch scoring for applications; probability scores usable as an automated risk signal.
Banks and credit agencies	Loan-amount recommendations from the maximum safe EMI; portfolio screening; documented, repeatable decisions with MLflow-versioned models.

Stakeholder	Impact
Loan officers and underwriters	A financial-health snapshot in seconds; a clear recommendation with confidence; time freed for complex cases.
Applicants	Transparent affordability picture, including headroom against the requested EMI, and counter-offers rather than flat rejections.

6. Recommendations for financial institutions

1. Adopt a tiered decision workflow. Auto-approve Eligible predictions with high confidence, auto-decline high-confidence Not_Eligible, and route High_Risk and low-confidence cases to an underwriter. Set the confidence thresholds with your own loss data.
2. Run in shadow mode first. Score live applications alongside the existing process for one to two months and compare decisions before any automation.
3. Use the maximum safe EMI for counter-offers. When a request exceeds the safe EMI, propose a smaller amount or longer tenure instead of rejecting.
4. Price High_Risk applicants, do not just decline them. The class is defined as marginal, so higher rates or added security (a co-applicant or guarantor) can turn it into a controlled opportunity.
5. Set scenario-specific policies. Vehicle and personal loans have the lowest eligibility, so use stricter documentation or co-applicant rules there, and pre-approval campaigns for the higher-eligibility e-commerce and home-appliance segments.
6. Coach applicants on the drivers that matter. Existing loan obligations and the EMI-to-disposable-income ratio matter most, so advise customers to reduce existing EMIs before applying.
7. Document every decision for compliance. Store inputs, class probabilities, model version and feature importance for each decision; consider adding SHAP explanations for per-applicant reasons.
8. Audit for fairness. Test approval rates across gender, marital status and age before deployment, even though these features rank low in importance.
9. Monitor and retrain. Track accuracy, drift in salary and credit-score distributions, and the approval-error rate; use the MLflow model registry to version, compare and roll back models.

7. Risks and limitations

- Synthetic-style data: the dataset follows domain rules and the maximum-EMI model depends heavily on one feature (existing loans, around 65% importance). Real portfolios will differ, so recalibrate and re-validate on real data before relying on the scores.
- Class mix: about 77% of records are Not_Eligible and only about 4% High_Risk. Real lending books may be far less skewed.
- Borderline class: High_Risk is the hardest class to predict, which is why it should stay a human-review bucket.
- Demo-grade administration: the deployed app's file system is temporary, so records created or edited in the Admin page may reset when the app restarts. A production system needs a real database, authentication and an audit log.
- Regulation: automated credit decisions are subject to local rules (for example RBI guidelines in India); a legal and compliance review is needed before use in live lending.
- Interest not modelled: the requested EMI shown in the app is an approximation without interest, so final pricing must come from the institution's own rate engine.

8. Suggested roadmap

Phase	Timeframe	Goal
1. Pilot (shadow mode)	Months 1–2	Compare model against underwriter decisions; measure agreement and error cost
2. Assisted decisions	Months 3–4	Underwriters use predictions as a recommendation; collect feedback
3. Partial automation	Months 5–6	Auto-decide high-confidence Eligible and Not_Eligible; keep human review for the rest
4. Scale and monitor	Ongoing	Production database, authentication, monitoring dashboard, scheduled retraining

9. Conclusion

EMIPredict AI shows that a data-driven approach can make EMI decisions faster, more consistent and easier to explain. Its strongest value is not replacing underwriters but focusing them on the small share of genuinely borderline applications, while giving every applicant an instant, transparent view of what they can afford. With shadow-mode validation, fairness checks and a production-grade backend, the platform is a sound foundation for high-volume lending.

This is an educational project. Its predictions are not a substitute for real lending decisions.